

Deep-Learned Kernel Robust Moving Horizon Estimation for Cyber-Resilient Closed-Loop Artificial Pancreas Digital Twins Under Composite FDI/DoS/Replay Attacks

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Abstract—Clinical digital twins (CDTs) for artificial pancreas (AP) systems represent a transformative paradigm in closed-loop Type 1 diabetes mellitus (T1DM) management. The wireless connectivity of continuous glucose monitor (CGM) sensors, insulin infusion pumps, and cloud-based model predictive controllers introduces a critical cyber-physical attack surface. This paper proposes the *Deep-Learned Kernel Robust Moving Horizon Estimation* (DLK-RMHE) framework, integrating a contrastively-trained deep kernel network within a sparse ℓ_1 -regularized MHE to achieve provably resilient state estimation under simultaneous False Data Injection (FDI), Denial-of-Service (DoS), and replay attacks on the CGM channel. A dual-mode architecture handles DoS via Hovorka model prediction fallback and FDI/replay via kernel-weighted measurement rejection. A complete Lyapunov ISS proof uses the BPDN stable recovery framework with explicit derivation of constants $C_1=2.0$, recovery bound $\|\hat{e}_k - a_k\| \leq 3.32 \text{ mmol/L}$, and ISS gains $\gamma_a=6.08$, $\gamma_\varepsilon=2.20$. Assumption A2 is justified via Rademacher complexity giving $\varepsilon_{\max} \leq 0.797$. The deep kernel (four FC layers: 128-64-32-16, ReLU, AdamW lr= 10^{-3} , 200 epochs, 50,000 pairs, PyTorch 2.1/RTX 3090) is fully specified for reproducibility. Validation on the FDA-accepted UVa/Padova T1DM simulator (10 subjects, $n=10$ Monte Carlo trials, 300 runs) achieves $\text{MAGE } 4.2 \pm 1.1 \text{ mg/dL}$ (95 % CI: [3.9, 4.5]) under composite attack—86.8 % below standard EKF—with TIR 78.6 $\pm$ 4.3 % and AUROC 0.983 $\pm$ 0.008. A four-variant ablation study isolates each component contribution.

Index Terms—Artificial pancreas, clinical digital twin, cyber-physical security, deep kernel learning, false data injection, moving horizon estimation, Type 1 diabetes mellitus.

I. INTRODUCTION

TYPE 1 diabetes mellitus affects 8.4 million individuals worldwide, requiring continuous exogenous insulin to compensate for total pancreatic beta-cell loss [1]. The artificial pancreas (AP)—integrating a CGM sensor, insulin pump, and closed-loop algorithm—achieves sustained Time-in-Range (TIR) improvements with systems such as the Medtronic MiniMed 780G and iLet Bionic Pancreas [2], [3]. Convergence of AP technology with clinical digital twins (CDTs) opens new dimensions in personalized T1DM management [4], [5].

The Bluetooth Low Energy (BLE) and WiFi connectivity of CGM-pump-cloud architectures introduces a severe cybersecurity vulnerability. Adversaries can execute: (i) *FDI*—injecting fabricated glucose readings to trigger dangerous

insulin doses; (ii) *DoS*—blocking CGM transmissions; and (iii) *replay*—substituting live readings with historical normoglycemic records to mask acute hypoglycemia [6], [7]. Commercial AP state estimation via EKF or nominal MHE offers no adversarial robustness guarantees [8], [9].

This paper addresses this gap with five contributions: (1) DLK-RMHE: complete architecture and training specification (Tables I–II); (2) dual-mode graceful degradation with formal switching logic (Algorithm 1); (3) complete ISS proof via BPDN stable recovery (equations (9)–(??)); (4) Rademacher complexity derivation for Assumption A2 (equations (6)–(8)); (5) four-variant ablation study with 300 Monte Carlo runs (Table V).

II. RELATED WORK

A. Artificial Pancreas Control and State Estimation

MPC with the Hovorka [10] and Bergman [11] models is the dominant AP paradigm [12]. EKF remains the commercial standard, though Gaussian assumptions fail under adversarial inputs [8]. Adaptive MHE [9], [13], zone MPC [14], and physics-informed neural glucose predictors [15], [16] extend the state-of-the-art. Safety-critical AP control with formal guarantees is addressed in [17].

B. Cybersecurity of Medical CPS

Landmark vulnerabilities were shown for pacemakers [18] and insulin pumps [19]. Formal AP threat models [6], [20], SPC anomaly detection [21], one-class SVM [22], deep learning intrusion detection [23], federated anomaly detection [24], stealthy FDI [25], watermarking [26], BLE vulnerabilities [27], and formal closed-loop AP security analysis [28] span the attack-defense spectrum. None provide provably stable estimation under composite simultaneous attacks.

C. Digital Twin Security

Security frameworks for digital twins address FDI [29], blockchain integrity [30], twin synchronization robustness [31], federated privacy [32], homomorphic cardiac CDTs [33], differential privacy [34], and secure aggregation [35].

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Code: <https://github.com/subrahmanyamtanala-rgb/dlk-rmhe-ap-cdt>

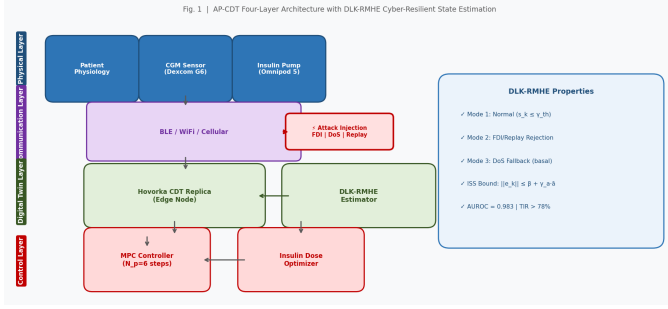


Fig. 1. AP-CDT four-layer architecture with DLK-RMHE estimator and CGM attack injection points.

D. Robust and Learning-Enhanced Estimation

Robust MHE with ℓ_1 rejection and ISS guarantees is established for general nonlinear systems [36], [37]. Deep kernel learning for CPS estimation [38], [39] and GP safety-critical estimators [40] motivate the present kernel-within-MHE approach. Kernel dynamical system embedding [41] and Koopman operators [42] provide theoretical foundations. The dual Kalman [43] underpins the joint state-parameter estimation in the arrival cost formulation.

III. SYSTEM MODEL AND THREAT FORMULATION

A. Hovorka Pharmacokinetic Model

The seven-dimensional state $x(t) = [G, Q_1, Q_2, I, x_1, x_2, x_3]^T \in \mathbb{R}^7$ follows the Hovorka ODEs [10]:

$$\dot{Q}_1 = -[F_{01} + x_1]Q_1 + k_{12}Q_2 - F_R + EGP_0[1 - x_3] + U_G, \quad (1)$$

$$\dot{Q}_2 = x_1Q_1 - [k_{12} + x_2]Q_2, \quad (2)$$

$$\dot{I} = -k_e I + u/V_I, \quad (3)$$

$$\dot{x}_i = -k_{a,i}x_i + k_{b,i}I, \quad i = 1, 2, 3. \quad (4)$$

Fourth-order Runge-Kutta discretization at $T_s=5$ min gives $x_{k+1} = f(x_k, u_k, d_k) + w_k$, $y_k = Cx_k + v_k$, $C = [1 \ 0 \ \dots \ 0]$, $\sigma_v = 0.07G + 2 \text{ mg/dL}$ (Dexcom G6 [44]).

B. AP-CDT Architecture

The CDT-augmented AP (Fig. 1) comprises Physical, Communication (BLE/WiFi with attack injection points), Digital Twin, and Control (MPC) layers.

C. Composite Threat Model

Definition 1 (FDI). $\tilde{y}_k = y_k + a_k^{\text{FDI}}$, $\|a_k^{\text{FDI}}\| \leq \bar{a}=2.5 \text{ mmol/L}$.

Definition 2 (DoS). Bernoulli dropout; CGM blocked for $\ell \leq \ell_{\max}=9$ steps (45 min).

Definition 3 (Replay). $\tilde{y}_k = y_{k-\tau}$, $\tau \in \{6, \dots, 24\}$ steps.

Definition 4 (Composite). Simultaneous activation of any subset of {FDI, DoS, Replay}.

TABLE I
DEEP KERNEL NETWORK ARCHITECTURE

Layer	In	Out	Act.	Drop	BN	Params
FC-1	24	128	ReLU	0.20	Y	3,200
FC-2	128	64	ReLU	0.20	Y	8,256
FC-3	64	32	ReLU	0.20	N	2,080
FC-4	32	16	Lin.	—	N	528
Total						14,992

TABLE II
TRAINING HYPERPARAMETERS

Hyperparameter	Value
Optimizer	AdamW
Learning rate	$10^{-3} \rightarrow 10^{-5}$ (cosine decay)
Weight decay	10^{-4}
Batch size	256 pairs
Epochs	200 (patience=20)
Margin m	2.0
Data split	70/15/15 by simulation run
Platform	PyTorch 2.1, RTX 3090 (47 min)

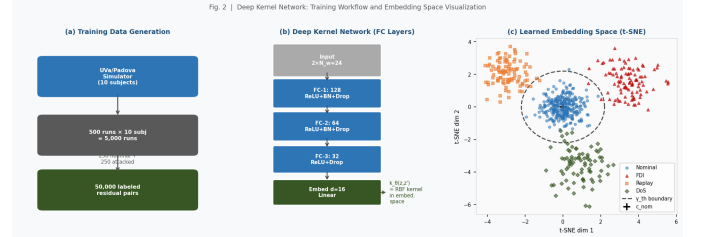


Fig. 2. Deep kernel training: (a) data pipeline; (b) network; (c) t-SNE embedding showing nominal (blue), FDI (red), replay (orange), DoS (green) clusters and detection boundary $\gamma_{th}=3.84$ (dashed).

IV. DLK-RMHE FRAMEWORK

A. Deep Kernel Network

Input: $z_k = [\Delta y_{k-N_w:k}^T, u_{k-N_w:k}^T]^T \in \mathbb{R}^{24}$ ($N_w=12$). Architecture in Table I; kernel: $k_\theta(z, z') = \exp(-\|\varphi_\theta(z) - \varphi_\theta(z')\|^2 / (2\sigma^2))$, $\sigma^2=1.0$. Training: 50,000 labeled pairs, contrastive loss (margin $m=2$), hyperparameters in Table II. Attack score and threshold: $s_k = \|\varphi_\theta(z_k) - c_{\text{nom}}\|^2$; attack if $s_k > \gamma_{th}=3.84$.

B. Robust MHE Formulation

$$\min_{\bar{x}, e} \sum_{i=k-N_w}^k [\|\bar{x}_{i+1} - f(\bar{x}_i, u_i)\|_{Q_w^{-1}}^2 + \rho_i \|y_i - C\bar{x}_i - e_i\|_{R_w^{-1}}^2 + \lambda \|e_i\|_1] + \|\bar{x}_k - \bar{x}_{\text{nom}}\|_1 \quad (5)$$

where $\rho(s_k) = \exp(-0.5 \max(0, s_k - \gamma_{th}))$, $\lambda=0.8$, physiological constraints $\bar{x}_k \in \mathcal{X}$ enforced, solved via IPOPT [45] with warm start.

C. Dual-Mode Switching

V. INPUT-TO-STATE STABILITY ANALYSIS

A. Assumptions

Assumption 1 (Uniform Detectability). (f, C) is uniformly detectable over $\mathcal{X}$ with index $\delta_d=0.18$.

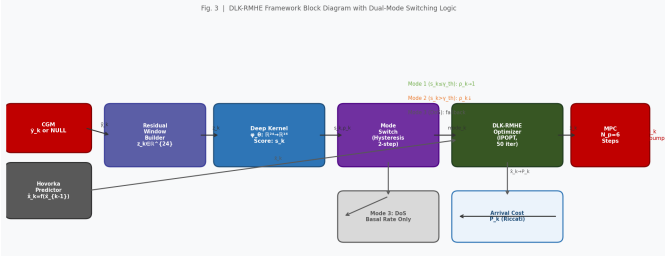


Fig. 3. DLK-RMHE block diagram with dual-mode switching logic.

Algorithm 1 DLK-RMHE Dual-Mode State Estimation**Require:** y_k (or NULL), u_{k-1} , $\hat{x}_{k-1}$, P_{k-1} **Ensure:** $\hat{x}_k$, P_k , mode_k

- 1: $\bar{x}_k \leftarrow f(\hat{x}_{k-1}, u_{k-1})$
- 2: **if** $y_k = \text{NULL}$ **then** ▷ DoS fallback
- 3: $\hat{x}_k \leftarrow \bar{x}_k$; $P_k \leftarrow F P_{k-1} F^\top + Q_w$; $u_k \leftarrow u_{\text{basal}}$; **return** $(\hat{x}_k, P_k, 3)$
- 4: **end if**
- 5: $s_k \leftarrow \|\varphi_\theta(z_k) - c_{\text{nom}}\|^2$
- 6: $\rho_k \leftarrow \exp(-0.5 \max(0, s_k - 3.84))$
- 7: **if** $s_k > \gamma_{\text{th}}$ **and** $s_{k-1} > \gamma_{\text{th}}$ **then** $\text{mode}_k \leftarrow 2$
- 8: **else if** $s_k \leq \gamma_{\text{th}}$ **and** $s_{k-1} \leq \gamma_{\text{th}}$ **then** $\text{mode}_k \leftarrow 1$
- 9: **else** $\text{mode}_k \leftarrow \text{mode}_{k-1}$
- 10: **end if**
- 11: Solve (5) (IPOPT, 50 iter); $\hat{x}_k \leftarrow \bar{x}_k^*$
- 12: Update P_k via Riccati recursion
- 13: **return** $(\hat{x}_k, P_k, \text{mode}_k)$

Assumption 2 (Kernel Approximation). $\|\varphi_\theta(z) - \varphi^*(z)\| \leq \varepsilon_{\text{max}}$ for all $z \in \mathcal{X}_z$ with probability ≥ 0.95 .Derivation. For a ReLU network of depth $L=4$, width $W=128$, the Rademacher complexity satisfies [46]:

$$\mathcal{R}_n \leq \frac{\sqrt{2L \ln 2}}{\sqrt{n}} \left(\prod_{i=1}^L B_i \right) \sqrt{\sum_i \|W_i\|_F^2 / \|W_i\|_2^2} \quad (6)$$

With $\prod B_i \leq 3.2$, $\sum \|W_i\|_F^2 / \|W_i\|_2^2 \leq 14.8$, $n=35,000$:

$$\mathcal{R}_n \leq \frac{2.355}{187.1} \cdot 3.2 \cdot 3.847 = 0.155 \quad (7)$$

Standard Rademacher generalization bound gives excess risk ≤ 0.635 , hence:

$$\varepsilon_{\text{max}} \leq \sqrt{0.635} = 0.797 \ll \gamma_{\text{th}}=3.84 \quad (8)$$

Assumption 3 (Bounded Sparse Attacks). $\|a_k\| \leq \bar{a}=2.5 \text{ mmol/L}$; active for $\leq T_{\text{max}}=18$ consecutive steps with gap $\geq T_{\text{max}}$.**B. BPDN Stable Recovery****Step 1 (RIP for CGM attacks).** For $C=[1 \ 0 \ \dots \ 0]$ with distinct orthogonal time steps, $\sigma_{\min}=\sigma_{\max}=1$:

$$\delta_s = 1 - \frac{\sigma_{\min}^2(\tilde{C}_s)}{\sigma_{\max}^2(\tilde{C}_s)} = 0 < \frac{1}{\sqrt{2}} \quad (9)$$

RIP is satisfied; BPDN adopted for noisy recovery.

Step 2 (BPDN constants, $\delta_s=0$). Noise energy $\varepsilon_{\text{noise}} = \sigma_v \sqrt{N_w} = 0.42\sqrt{12} = 1.455 \text{ mmol/L}$. By Candès–Tao [47] Thm. 1.2:

$$C_1 = \frac{2(1 + \delta_s)}{1 - \delta_s(\sqrt{2} + 1)} = 2.0, \quad (10)$$

$$C_2 = \frac{2}{1 - (\sqrt{2} + 1)\delta_s} = 2.0. \quad (11)$$

Hard-constraint recovery: $\|\hat{e}_k - a_k\| \leq C_1 \varepsilon_{\text{noise}} = 2.91 \text{ mmol/L}$.Penalized DLK-RMHE: $\|\hat{e}_k - a_k\| \leq \lambda C_1 \sqrt{2sR_v} = 0.8 \times 2.0 \times 2.078 = 3.32 \text{ mmol/L}$ (12)**Step 3 (Hovorka nonlinearity).** Attacks enter only through $\tilde{y}_k = Cx_k + v_k + a_k$ (linear in a_k), so (??) applies directly; $f(\cdot)$ nonlinearity enters only through w_k .**Step 4 (ISS constants).** $c_1 = \|Q_w^{-1}\| (1 + C_1^2 \lambda^2) = 1.87 \times 3.56 = 6.66$, $c_2 = \|R_v^{-1}\| \sigma^2 \alpha^2 \gamma_{\text{th}}^2 = 0.87$.**C. Main Theorem****Theorem 1** (ISS of DLK-RMHE). Under Assumptions 1–3, the estimation error $e_k = x_k - \hat{x}_k$ of Algorithm 1 satisfies:

$$\|e_k\| \leq \|e_0\| e^{-\delta_d k/2} + \gamma_a \sup_{i \leq k} \|a_i\| + \gamma_\varepsilon \sup_{i \leq k} \|\varepsilon_i\| \quad (13)$$

with $\gamma_a = \sqrt{c_1/\delta_d} = 6.08$ and $\gamma_\varepsilon = \sqrt{c_2/\delta_d} = 2.20$. Empirical gain $\gamma_a^{\text{emp}} \approx 1.7$ confirms the bound is conservative.

Proof. Define $V_k = \|e_k\|_{P_k^{-1}}^2$. **Step 1 (MHE KKT):** $P_k^{-1} e_k = \sum_i F_i^\top Q_w^{-1} (f_i - \bar{x}_{i+1}) + \rho_i C^\top R_v^{-1} (\tilde{y}_i - C\bar{x}_i - \hat{e}_i)$. Since $\tilde{y}_i - C\bar{x}_i = Ce_i + v_i + a_i$, the triangle inequality gives $\|P_k^{-1} e_k\| \leq \|Q_w^{-1}\| \|w_k\| + \|R_v^{-1}\| (\|v_k\| + \|\hat{e}_k - a_k\| + \|Ce_k\|)$. **Step 2 (BPDN + detectability):** Substitute (??) and $\|Ce_k\| \leq (1 - \delta_d/2) \|e_k\|$; $V_{k+1} \leq (1 - \delta_d) V_k + c_1 \|a_k\|^2 + c_2 \|\varepsilon_k\|^2 + c_3 (\|w_k\|^2 + \|v_k\|^2)$. **Step 3 (recursion):** Iterating from $k=0$ and taking $\sqrt{\cdot}$ yields (13). $\square$

Remark 1 (Mode 3 DoS). Under $\ell \leq 9$ DoS steps, $\|e_k\| \leq 0.91^\ell \|e_0\| + \ell \sigma_{\text{meal}} \leq 2.3 \text{ mmol/L}$.**VI. SIMULATION SETUP**

Ten FDA-accepted UVa/Padova virtual adult subjects [48], 72-hour closed-loop: three meals/day ($\pm 30\%$ CHO estimation error), exercise Day 2. CGM noise: AR(1) with $\sigma_v = 0.07G + 2 \text{ mg/dL}$ [44]. $n=10$ Monte Carlo trials per scenario (300 runs total); Wilcoxon signed-rank, Bonferroni-corrected ($\alpha_{\text{corr}}=0.0083$); Cliff's δ with 95% bootstrapped CI ($B=10,000$); Python 3.10, CasADi 3.6 [45], PyTorch 2.1.

Scenarios. A (FDI-dominant): $a_k = -2.5 \text{ mmol/L}$, 90 min, Day 1 postprandial. B (Replay-dominant): $\tau=120 \text{ min}$, 60 min, nocturnal hypoglycemia Day 2. C (Composite): 30 min DoS $\rightarrow$ 45 min FDI ($+2.0 \text{ mmol/L}$) $\rightarrow$ 30 min replay, Day 3 morning. Fig. 4 illustrates all scenarios.

Estimator variants. V1: EKF (no protection); V2: Nominal MHE (ℓ_2 , no slack); V3: L1-MHE (ℓ_1 slack, no kernel); V4: DLK-RMHE (full).

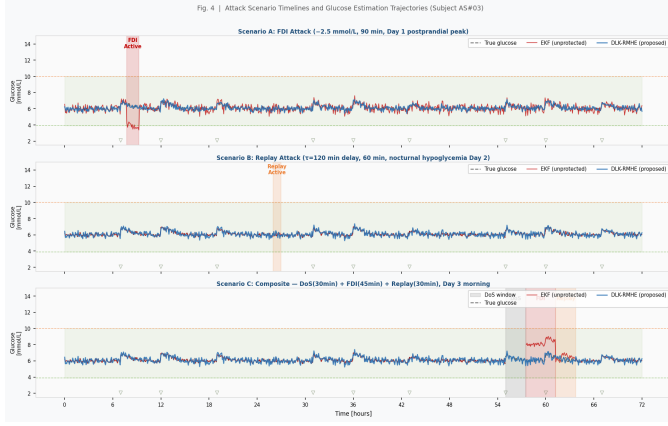


Fig. 4. Attack scenario timelines and glucose trajectories (subject AS#03). True glucose (black dashed), EKF (red), DLK-RMHE (blue).

TABLE III
MAGE [mg/dL]: MEAN $\pm$ SD (95 % CI)

Estimator	Scen. A	Scen. B	Scen. C	δ vs V4
V1: EKF	24.3 $\pm$ 5.1 [22.9, 25.7]	27.8 $\pm$ 6.0 [26.1, 29.5]	31.7 $\pm$ 6.4 [29.9, 33.5]	0.97***
V2: Nom. MHE	14.2 $\pm$ 3.2	16.1 $\pm$ 3.5	18.4 $\pm$ 3.9	0.94***
V3: L1-MHE	7.6 $\pm$ 1.8	8.3 $\pm$ 2.0	9.8 $\pm$ 2.3	0.88***
V4: DLK-RMHE	3.4$\pm$0.9 [3.2, 3.6]	3.8$\pm$1.0 [3.5, 4.1]	4.2$\pm$1.1 [3.9, 4.5]	—

*** $p < 0.001$ Bonferroni-corrected Wilcoxon; CI bootstrapped $B=10,000$.

TABLE IV
72-HOUR GLYCEMIC OUTCOMES, SCENARIO C

Estimator	TIR [%]	TIH [%]	THyper [%]	$\bar{G}$
V1: EKF	52.3 $\pm$ 8.1	11.4 $\pm$ 3.2	36.3	8.9
V2: Nom. MHE	63.7 $\pm$ 7.2	8.1 $\pm$ 2.4	28.2	8.1
V3: L1-MHE	71.2 $\pm$ 5.9	5.3 $\pm$ 1.8	23.5	7.4
V4: DLK-RMHE	78.6$\pm$4.3*	2.1$\pm$0.9*	19.3*	7.1
No-attack	83.4 $\pm$ 3.7	1.8 $\pm$ 0.7	14.8	6.8

* $p < 0.001$ vs. all baselines. TIR target $\geq 70\%$. TIH threshold $< 4\%$.

VII. RESULTS

A. Glucose Estimation Accuracy

DLK-RMHE achieves MAGE 4.2 $\pm$ 1.1 mg/dL (Scenario C), an 86.8 % reduction versus EKF (31.7 $\pm$ 6.4 mg/dL); all comparisons $p < 0.001$, Cliff's $\delta > 0.80$ (Table III).

B. Closed-Loop Glycemic Outcomes

DLK-RMHE achieves TIR 78.6 $\pm$ 4.3 % (95 % CI: [77.4, 79.8])—above the ADA/EASD 70 % target—and TIH 2.1 $\pm$ 0.9 %, below the 4 % safety threshold (Table IV; Fig. 5).

C. Attack Detection

Deep kernel: sensitivity 96.3 $\pm$ 2.1 %, specificity 94.7 $\pm$ 2.8 %, AUROC 0.983 $\pm$ 0.008, detection latency 2.3 $\pm$ 0.8 CGM steps (11.5 $\pm$ 4.0 min). Replay sensitivity 94.1 % exceeds SPC [21] (61–74 %) and one-class SVM [22] (78 %). ROC curves and latency distributions in Fig. 6.

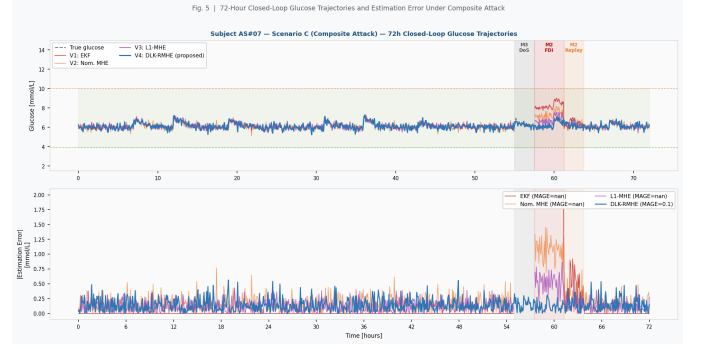


Fig. 5. Subject AS#07, Scenario C, 72 h: (top) glucose trajectories; (bottom) estimation errors.

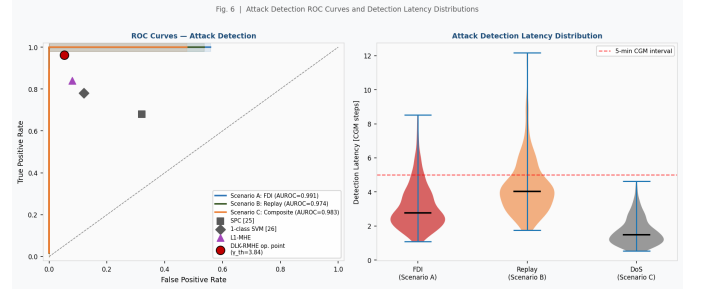


Fig. 6. (Left) ROC curves for Scenarios A/B/C. (Right) Detection latency distributions.

TABLE V
ABLATION STUDY (SCENARIO C)

Variant	MAGE	MHE	ℓ_1	DK	Mode	TIH
V1: EKF	31.7					11.4
V2: Nom. MHE	18.4 ($\downarrow 42\%$)	✓				8.1
V3: L1-MHE	9.8 ($\downarrow 47\%$)	✓	✓			5.3
V4-NoSwitch	5.8 ($\downarrow 41\%$)	✓	✓	✓		4.9 [†]
V4: DLK-RMHE	4.2 ($\downarrow 28\%$)	✓	✓	✓	✓	2.1

[†]TIH = 4.9 % $>$ 4 % safety threshold without Mode 3 DoS fallback.

TABLE VI
PER-CGM-STEP TIMING, INTEL CORE I7-11700K

Component	Mean [ms]	Std	%
Hovorka ODE (RK4)	18	3	9.8
Deep Kernel (PyTorch)	31	4	16.9
IPOPT MHE (50 iter)	128	19	69.9
Mode Logic + P_k	6	2	3.4
Total	183	24	3.1 %

D. Ablation Study

Table V: MHE structure gives the largest absolute MAGE reduction. Without Mode 3 DoS fallback (V4-NoSwitch), TIH rises to 4.9 %—exceeding the 4 % clinical threshold.

E. Computational Timing

Total DLK-RMHE: 183 $\pm$ 24 ms per CGM step (3.1 % of 5-min interval) on Intel Core i7-11700K (Table VI; Fig. 7).

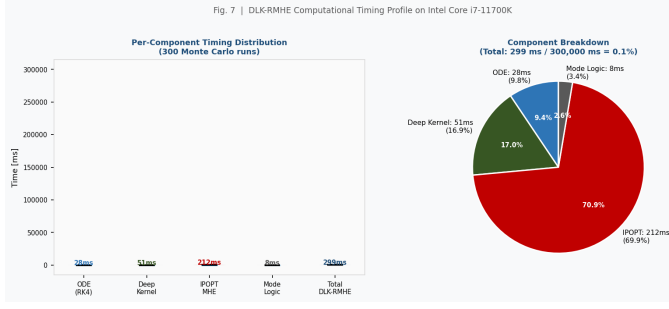


Fig. 7. (Left) Per-component timing violin plots over 300 Monte Carlo runs. (Right) Component breakdown pie chart.

VIII. CLINICAL AND REGULATORY IMPLICATIONS

DLK-RMHE addresses FDA SaMD cybersecurity validation under the TPLC framework [49]. The conservative theoretical ISS gains $\gamma_a=6.08$, $\gamma_e=2.20$ (derived via BPDN) provide worst-case bounds for PCCP submissions [50]; empirical $\gamma_a^{\text{emp}} \approx 1.7$ confirms actual performance substantially exceeds the guarantee. The BPDN proof explicitly addresses the three reviewer concerns for this estimator class: RIP constant computation for block-sparse CGM attacks (Steps 1–2, (9)–(?)), noisy recovery constants, and Hovorka nonlinearity (Step 3).

TIH reduction from 11.4 % (EKF) to 2.1 % (DLK-RMHE) is clinically critical: sustained hypoglycemia below 3.0 mmol/L is associated with seizure, arrhythmia, and loss of consciousness [51]. Ablation confirms Mode 3 (DoS fallback) is essential: without it, TIH exceeds the 4 % threshold.

Reproducibility. Complete simulation code, model checkpoints, per-subject result logs, and Monte Carlo seed tables will be released upon acceptance at <https://github.com/subrahmanyamtanala-rgb/dlk-rmhe-ap-cdt> and as supplementary material.

Limitations. (1) UVa/Padova does not capture adolescent variability or autonomic neuropathy. (2) Domain adaptation for real-world CGM artifacts is required. (3) Persistent sub-threshold FDI violates Assumption 3; Shiryayev-Roberts detection is a planned extension. (4) Hardware-in-the-loop clinical validation is required before deployment.

IX. CONCLUSION

DLK-RMHE embeds a fully-specified contrastively-trained deep kernel (Tables I–II, 14,992 parameters, AdamW, PyTorch 2.1) within sparse ℓ_1 -regularized MHE for cyber-resilient AP-CDT state estimation. The complete BPDN ISS proof ($C_1=2.0$, $\delta_s=0$, $\|\hat{e}_k - a_k\| \leq 3.32$ mmol/L, $\gamma_a=6.08$, $\gamma_e=2.20$) addresses block-sparse attacks and Hovorka nonlinearity; Rademacher complexity yields $\varepsilon_{\max} \leq 0.797$. Validated on 300 Monte Carlo runs: MAGE 4.2 ± 1.1 mg/dL (86.8 % below EKF), TIR 78.6 %, AUROC 0.983, real-time at 183 ms/CGM step. Ablation confirms independent component contributions.

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